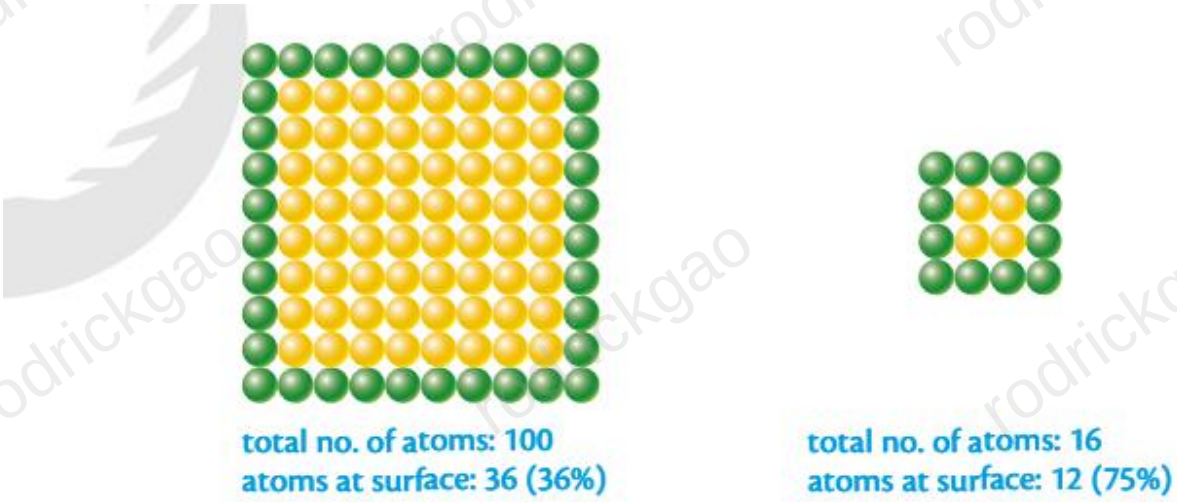
E203 Probing into the Nano World

E2.3.1 · E2.3.3 Nanoscale materials

A nanometre is one billionth of a metre. Nanomaterials usually have at least one dimension around 1–100 nm. Increased surface-atom fractions and structural or quantum effects can change their electrical, mechanical and optical properties relative to bulk material. Do not attribute every change to surface area alone; explain the material-specific mechanism.

$$1\ \text{nm} = 10^{-9}\ \text{m}$$

Nanomaterials typically have at least one dimension between 1 and 100 nm.

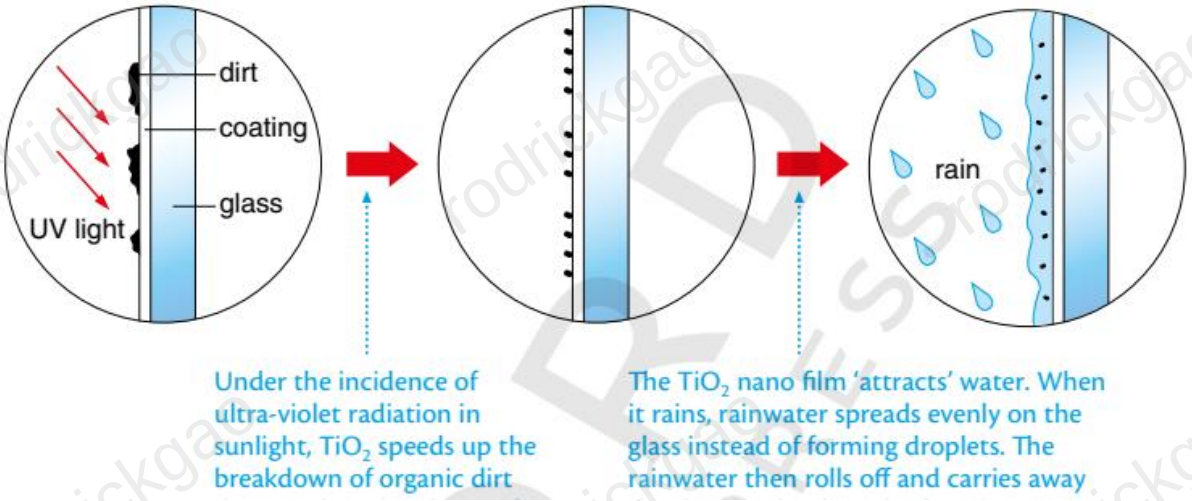


Smaller particles have a greater fraction of atoms at their surfaces.

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E2.3.2 Self-cleaning coatings and carbon structures

Titanium-dioxide coatings combine ultraviolet-assisted breakdown of organic dirt with a hydrophilic surface that lets rain spread and wash dirt away, unlike the lotus effect's water-repelling droplets. Carbon illustrates structure-property relationships: diamond has a three-dimensional covalent network, graphite has sliding layers, graphene is one honeycomb sheet, and nanotubes resemble rolled sheets. Different atomic arrangements produce different mechanical and electrical properties despite the same element.



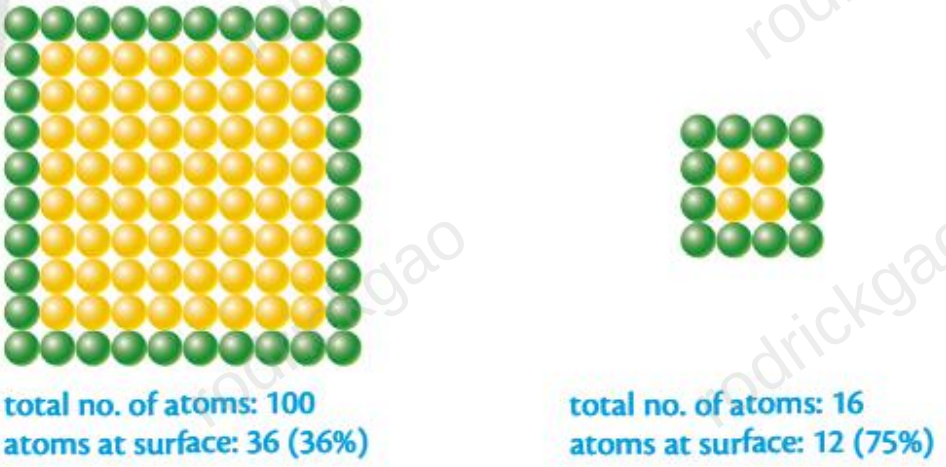
Hydrophilic self-cleaning by a titanium-dioxide coating.

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E2.3.4 Surface effects and the lotus effect

Reducing size increases surface-area-to-volume ratio and the fraction of surface atoms available to interact with the surroundings. A lotus leaf's nanoscale surface structures repel water, allowing rolling

droplets to carry dirt away. These are distinct textbook mechanisms; not all nanoscale properties have one common cause.



Smaller particles have a greater fraction of atoms at their surfaces.



The water-repelling lotus surface lets droplets roll away carrying dirt.

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E2.3.5 · E2.3.6 Rayleigh's criterion and resolving power

A finite aperture forms a diffraction pattern for each point. Rayleigh's limit occurs when one central maximum coincides with the other's first minimum. For a circular aperture, shorter wavelength or greater diameter reduces the minimum resolvable angle, improving resolution. Resolving two nearby points differs from merely magnifying their image.

$$\theta_R \approx 1.22\frac{\lambda}{d}$$

Rayleigh criterion for a circular aperture. θ_R: minimum resolvable angular separation (rad); λ: wavelength (m); d: aperture diameter (m), or pupil diameter for the eye.

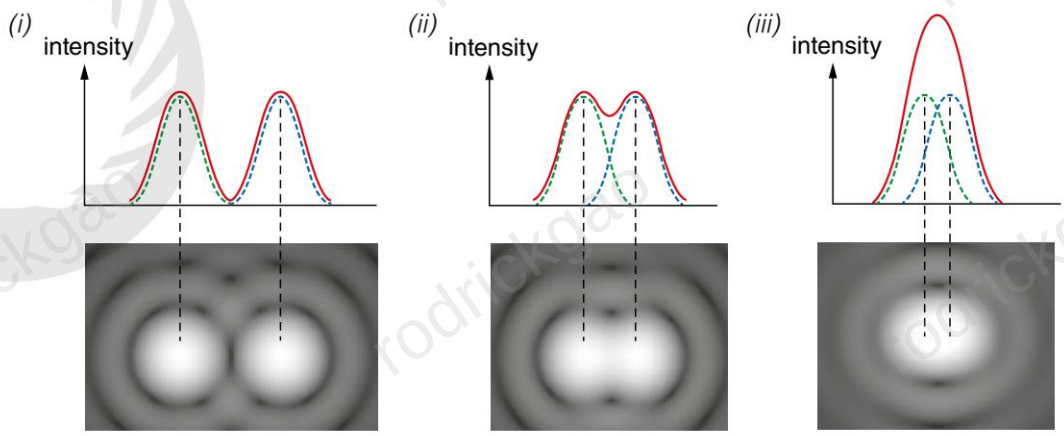


Fig 3.2d *According to Rayleigh criterion, two objects are (i) resolved clearly, (ii) just resolved, and (iii) not resolved.*

Compare clearly resolved, just resolved and unresolved pairs. At the Rayleigh limit, one central maximum falls on the first minimum of the other pattern.

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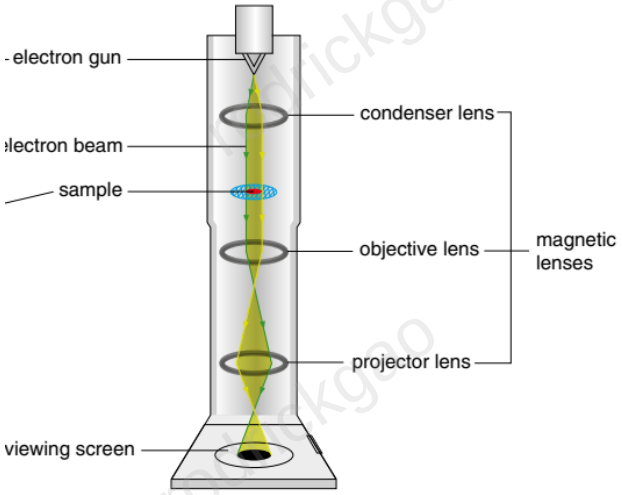
E2.3.7 · E2.3.8 Transmission electron microscopy

E2.3.7 Components of a transmission electron microscope

An electron gun produces the beam, condenser magnetic lenses focus it onto a thin sample, and objective and projector lenses form an enlarged transmitted-electron image on a screen or camera. Differences in sample transmission create contrast. Magnetic lenses bend electron paths, so identify each component's role in beam-travel order.

E2.3.8 Transmission electron microscopy

A transmission electron microscope passes fast electrons through a thin sample. Transmission differences create contrast, and magnetic lenses focus and magnify the beam. Higher accelerating voltage increases momentum and reduces de Broglie wavelength, improving the wavelength-related resolution limit. Actual resolution is also limited by lens and specimen effects.

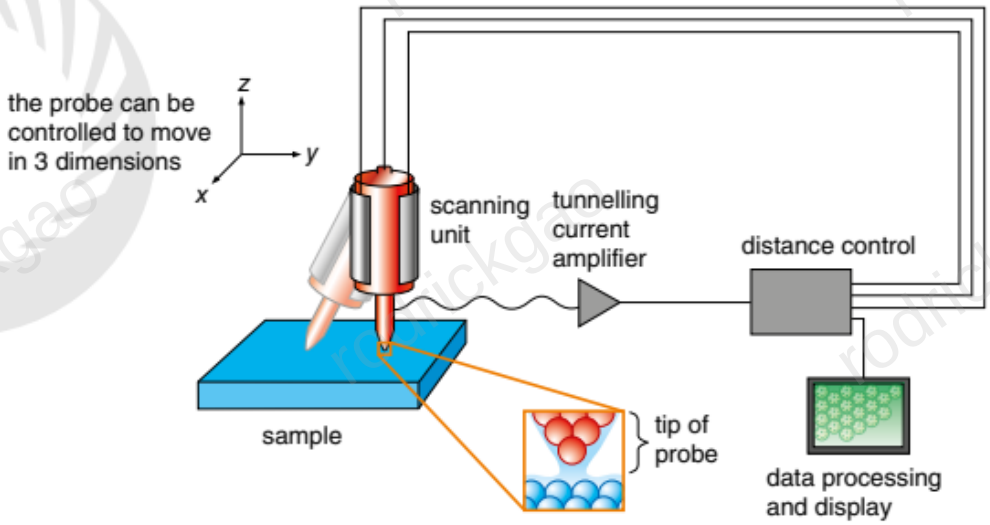


Follow the electron beam through the gun, magnetic lenses and sample to the imaging screen.

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E2.3.9 Scanning tunnelling microscopy

A conducting tip near a conducting sample detects a tunnelling current highly sensitive to separation. Scanning at fixed height while recording current, or adjusting height to keep current constant, reveals surface detail. The image is reconstructed from scanning and tunnelling signals, not an optical-lens photograph of atoms.



A scanning tip detects tunnelling current across a very small gap to obtain surface information.

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